

Quantum Field Theory

We're now going to pivot to introducing quantum field theory — the theoretical framework underlying all of particle physics, which we'll use extensively in calculations for the rest of the course.

After some motivation, we'll review classical field theory and discuss symmetries from a field theory perspective.

We'll then turn to quantization of scalar fields, using canonical quantization.

We won't have time to discuss path integrals — for that you'll have to consult a real field theory class.

This section will deal almost entirely with free fields.

We'll then turn our attention to interacting fields in perturbation theory, which leads naturally to scattering processes (and back to particle physics).

But why quantum fields?

A field is just a function of spacetime — it gives a value at each point $x^\mu = (\vec{x}, t)$.

This generalizes our classical / quantum mechanics where the degree of freedom q is just a function of t .

In fact, we'll see that QFT is essentially the QM of an infinite number of d.o.f's, living in 3+1 rather than 1d.

But why is this necessary? In the particle physics context, this is ~~because~~ necessitated by special relativity.

Some (hand-wavy) arguments:

- Antiparticles : The study of relativistic wave equations (Klein-Gordon, Dirac, Maxwell) reveals the presence of "negative energy" solutions, which seem problematic in a fundamental, quantum mechanical theory.

In QFT, these are naturally associated w/ (entirely non-problematic) anti-particle states.

- Particle creation / destruction :

QFT makes exchanging energy for particles via $E = \sqrt{p^2 + m^2}$ very simple, due to the infinite d.o.f.

→ This is otherwise very cumbersome in QM.

- Causality - we do not want signals travelling faster than the speed of light.

In ordinary QM, the amplitude for a particle propagating b/w spacelike separated points is nonzero. In QFT, this amplitude is (essentially) cancelled by the antiparticle propagation. This ensures causality is preserved.

- We already have fields! - the $\vec{E}$ & $\vec{B}$ fields in a fundamental theory are dynamical so it's essential to work in a context where we quantize these. In fact, these are a much richer, more complicated type of field we'll return to later on, called a gauge field.

With that motivation, we turn to classical field theory.

Classical Lagrangian Field Theory

Recall for a single d.o.f, $q(t)$, we wrote the action S as the time integral of the Lagrangian L :

$$S = \int dt L\left(q, \frac{dq}{dt}\right) = \int dt \left[\frac{1}{2} \left(\frac{dq}{dt}\right)^2 - V(q) \right]$$

We then extremized the action,

$$\begin{aligned} \delta S = 0 &= \int dt \left[\left(\frac{dq}{dt}\right) \frac{d}{dt} \delta q - \frac{dV}{dq} \delta q \right] \\ &= - \int dt \left[\frac{d^2 q}{dt^2} + \frac{dV}{dq} \right] \delta q \quad (\text{dropping the bdy term}) \end{aligned}$$

to find the equations of motion describing the dynamics:

$$\frac{d^2 q}{dt^2} + \frac{dV}{dq} = 0.$$

Now we generalize this by replacing $q(t)$ with a scalar field $\phi(x^\mu) = \phi(t, \vec{x}) \equiv \phi(x)$.

Scalar here means that at each spacetime point, ϕ just returns a single c-number.

The integration measure must also be generalized:

$$dt \rightarrow d^4 x = dt d^3 \vec{x} = dt dx dy dz$$

and the Lagrangian becomes a functional of the scalar field and its gradients.

Since we want a relativistic theory, we'll want the (scalar) Lagrangian to be invariant under Lorentz transformations. This suggests we write everything in terms of the four-gradient:

$$\frac{d}{dt} \rightarrow \frac{\partial}{\partial x^\mu} \equiv \partial_\mu = \left(\frac{\partial}{\partial t}, \vec{\nabla} \right)$$

With these replacements, the action of the scalar field is:

$$\begin{aligned} S &= \int d^4x \mathcal{L}[\phi, \partial_\mu \phi] \\ &= \int d^4x \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] \end{aligned}$$

where $\mathcal{L}$ is the Lagrangian density. (I will never remember to say density).

Note that both $\mathcal{L}$ and S are Poincaré invariant.

Now, just as for the generalized coordinate, we apply the principle of least action

— variational calculus of functions works the same as for ordinary coordinates.

We find:

$$\begin{aligned}
 0 = \delta S &= \int d^4x \left[g^{\mu\nu} \partial_\mu \phi \partial_\nu \delta\phi - \frac{dV}{d\phi} \delta\phi \right] \\
 &= - \int d^4x \delta\phi \left[g^{\mu\nu} \partial_\mu \partial_\nu \phi + \frac{dV}{d\phi} \right] \\
 &\quad + \int d^4x \partial_\nu \left[g^{\mu\nu} \partial_\mu \phi \delta\phi \right]
 \end{aligned}$$

The last term can be converted to a surface integral over the boundary of the four-volume using Stokes's Theorem.

Assuming $\delta\phi$ vanishes at $|x| \rightarrow \infty$, but is otherwise arbitrary, we obtain the eqn. of motion for the scalar field:

$$g^{\mu\nu} \partial_\mu \partial_\nu \phi + \frac{dV}{d\phi} = 0$$

$$\Rightarrow \square \phi + \frac{dV}{d\phi} = 0, \quad \text{where } \square = \partial_\mu \partial^\mu = g^{\mu\nu} \partial_\mu \partial_\nu$$

In 3-dimensional notation, this gives:

$$\frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi + \frac{dV}{d\phi} = 0$$

In the absence of a potential this is just a wave equation; yields the expected dynamics of an ordinary "field".

For most of this section, we'll work with the simplest allowed potential:

$$V(\phi) = \frac{1}{2} m^2 \phi^2$$

So far, m is just a parameter.

This will be the analogue of the harmonic oscillator — it is called free field theory — there are no interactions (which would come from terms $\sim \phi^3$ or higher)

We'll see soon that we are able to quantize the free theory exactly.

The equation of motion is:

$$\square \phi + m^2 \phi = 0.$$

This is known as the Klein-Gordon equation.

To interpret the parameter m , consider turning on a point source for the field (like adding a point charge in EM) — the source will ~~lead~~ set up a static $\phi(\vec{r})$ field, the solution to the equation:

$$-\nabla^2 \phi(r) + m^2 \phi(r) = -g \delta^3(\vec{r})$$

where g is a coupling constant.

The solution to this equation is:

$$\phi(\vec{r}) = \phi(r) = -\frac{g}{4\pi} \frac{1}{r} e^{-mr} \quad (r = |\vec{r}|)$$

→ The "force" associated with ϕ has a finite range.
 $R \sim m^{-1}$.

This is the familiar Yukawa potential, w/ m having the interpretation of a mass.

We'll later see this is exactly the right interpretation for many reasons.

Hamiltonian Field Theory

We'll soon quantize our field in the canonical formalism.

To do this, first we derive the Hamiltonian classically, then impose canonical commutation relations, with everything upgraded to operators.

First, define the "conjugate momentum density"

$$\pi(t, \vec{x}) = \frac{\delta \mathcal{L}}{\delta \dot{\phi}(t, \vec{x})} = \frac{\delta \mathcal{L}}{\delta \dot{\phi}(t, x)}$$

* the derivative

As always, the Hamiltonian (density) is the Legendre transform:

$$H = \int d^3x \mathcal{H} = \int d^3x [\pi(t, \vec{x}) \dot{\phi}(t, \vec{x}) - \mathcal{L}]$$

For the free theory, $\pi(t, \vec{x}) = \dot{\phi}(t, \vec{x})$, and

$$\begin{aligned} H &= \int d^3x \left[\underbrace{\pi \dot{\phi}}_{=\dot{\phi}^2} - \left(\underbrace{\frac{1}{2} \dot{\phi}^2}_{=\dot{\phi}^2} - \frac{1}{2} (\vec{\nabla} \phi)^2 - \frac{1}{2} m^2 \phi^2 \right) \right] \\ &= \int d^3x \left[\frac{1}{2} \dot{\phi}^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2 \right] \end{aligned}$$

The three terms in the Hamiltonian represent the energy densities associated with moving in time, steering in space, and in potential energy, respectively.

~~Summary, consequences~~

Symmetries, Conservation Laws & Noether's Theorem

In QFT, there is an intimate relationship between symmetries & conservation laws, encapsulated in a theorem by German ~~mathematician~~ ^{mathematical physicist} Emmy Noether.

Consider a continuous transformation of $\phi(x)$.

In infinitesimal form, we write

$$\phi(x) \rightarrow \phi'(x) = \phi(x) + \alpha \Delta\phi(x)$$

e. inf. parameter

w/ $\Delta\phi$ some deformation of the field configuration.

This transformation is a symmetry of the theory if it leaves the equations of motion invariant.

This is true if the action remains the same under the transformation, up to a surface term.

Thus, we require the Lagrangian to be invariant up to a '4-divergence'.

$$\mathcal{L}(x) \rightarrow \mathcal{L}(x) + \alpha \partial_\mu \mathcal{J}^\mu(x)$$

for some $\mathcal{J}^\mu(x)$.

Compare this to the result obtained by varying the fields.

$$\begin{aligned}
 \alpha \Delta \mathcal{L} &= \frac{\partial \mathcal{L}}{\partial \phi} (\alpha \Delta \phi) + \frac{\partial \mathcal{L}}{\partial (\partial_n \phi)} \partial_n (\alpha \Delta \phi) \\
 &= \alpha \partial_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_n \phi)} \Delta \phi \right] + \alpha \underbrace{\left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_n \frac{\partial \mathcal{L}}{\partial (\partial_n \phi)} \right]}_{=0 \text{ by eqns. of motion!}} \Delta \phi
 \end{aligned}$$

The first term can be identified as $\alpha \partial_n \mathcal{T}^n$:

$$\alpha \Delta \mathcal{L} = \alpha \partial_n \mathcal{T}^n = \alpha \partial_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_n \phi)} \Delta \phi \right]$$

Equivalently,

~~$$0 = \partial_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_n \phi)} \Delta \phi - \mathcal{T}^n \right]$$~~

$$0 = \partial_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_n \phi)} \Delta \phi - \mathcal{T}^n \right] \equiv \partial_n j^n$$

where we've defined a 4-current (density):

$$j^n(x) = \frac{\partial \mathcal{L}}{\partial (\partial_n \phi)} \Delta \phi - \mathcal{T}^n$$

We see that this current is conserved $\partial_n j^n = 0$;

A continuous symmetry (specifies $\Delta \phi$, $\mathcal{T}^n$) leads to a conservation law.

This conservation law can be expressed in terms of a conserved charge Q , defined as:

$$Q \equiv \int_{\text{all space}} d^3x j^0(x)$$

Since, (since $\partial_\mu j^\mu = 0 = \partial_0 j^0 - \partial_i j^i$)

$$\frac{dQ}{dt} = \int d^3x \frac{dj^0(x)}{dt} = - \int \vec{\nabla} \cdot \vec{j}(x) dx$$

$$= \int_{\text{boundary of space}} \vec{j}(x) \cdot d\vec{S} = 0$$

The last term vanishes assuming the current vanishes at the spatial boundary. The total charge Q is unchanged in time.

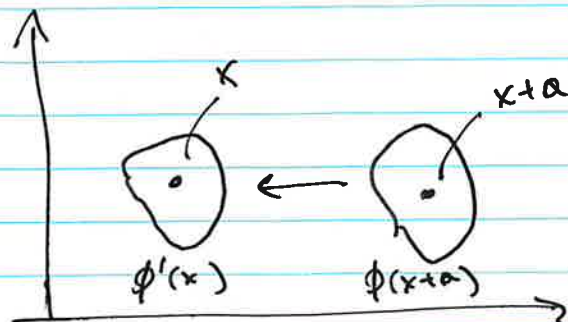
We can apply these ideas to the spacetime symmetries we just worked with.

Consider the translation: $x^\mu \rightarrow x^\mu + a^\mu$ ($a = \text{const.}$)

The scalar field is transformed as:

$$\phi(x) \rightarrow \phi'(x) = \phi(x+a) \simeq \phi(x) + \underbrace{a^\mu \partial_\mu \phi(x)}_{\equiv \Delta \phi}$$

This is an "active" transformation, in which we keep the coordinate system fixed and translate the field configuration!



The Lagrangian is also a scalar field, so it transforms in the same way!

$$\mathcal{L} \rightarrow \mathcal{L} + \alpha^\mu \partial_\mu \mathcal{L} = \mathcal{L} + \underbrace{\alpha^\nu \partial_\mu (\delta^\mu_\nu \mathcal{L})}_{\Delta \mathcal{L} = \alpha^\nu \partial_\mu (T^\mu_\nu)}$$

(note our infinitesimal parameter has a spacetime index
 $\rightarrow$ the "current" will have a separate one in addition to this)

Applying Noether's theorem, we identify

$$\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \Delta \phi = \alpha^\nu \left[\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \partial_\nu \phi \right], \quad T^\mu_\nu = \mathcal{L} \delta^\mu_\nu$$

So we get four, separately conserved currents for $\nu = 0, 1, 2, 3$.

These are conveniently grouped into a tensor known as the stress-energy tensor.

$$T^{\mu}_{\nu} \equiv \frac{\delta \mathcal{L}}{\delta(\partial_{\mu}\phi)} \partial_{\nu}\phi - \mathcal{L} \delta^{\mu}_{\nu}$$

which satisfies $\partial_{\mu} T^{\mu}_{\nu} = 0$.

Note that the conserved charge associated with time translations ($\nu=0$) is the Hamiltonian!

$$\begin{aligned} H &\equiv \int T^{00} d^3x = \int T^{\mu}_{\nu} g^{\nu 0} d^3x = \int T^{\mu}_0 d^3x \\ &= \int d^3x \left[\frac{\delta \mathcal{L}}{\delta \dot{\phi}} \dot{\phi} - \mathcal{L} \right] \equiv \int d^3x \mathcal{H}. \end{aligned}$$

The conserved charges associated w/ spatial translations are

$$\begin{aligned} P^i &\equiv \int T^{0i} d^3x = - \int T^{\mu}_i d^3x \\ &= - \int d^3x \frac{\delta \mathcal{L}}{\delta \dot{\phi}} \partial_i \phi = - \int d^3x \pi \partial_i \phi \end{aligned}$$

This can be interpreted as the physical momentum carried by the field.

We see that the spacetime symmetries of translations lead to energy & momentum conservation, respectively.

Besides spacetime symmetries, a theory may possess internal symmetries (ones not related to a change of space-time coordinates).

Consider a complex scalar field with Lagrangian:

$$\begin{aligned} \mathcal{L} &= |\partial_\mu \phi|^2 - m^2 |\phi|^2 \\ &= g^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi - m \phi^* \phi \end{aligned}$$

(Note the lack of $1/2$ factors — this is just convention for now, but you should think of it as counting the two separate "real" fields, $\phi = \phi_1 + i\phi_2$.)

This Lagrangian is invariant under the transformation

$$\phi(x) \rightarrow e^{i\alpha} \phi(x) \quad \text{for } \alpha = \text{const.}$$

This is a global $U(1)$ symmetry.

It is "global" in that α does not depend on x^μ , as opposed to a "local" transformation by $\alpha(x^\mu)$.

The $U(1)$ refers to the Lie group associated w/ this symmetry. A $U(1)$ symmetry is an Abelian symmetry group associated w/ rephasing of fields.

One can show that the conserved current in this theory is:

$$j^\mu(x) = i [(\partial^\mu \phi)^* \phi - \phi^* \partial^\mu \phi]$$

We'll have more to say about this later on.